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Decision Making for Two Samples

CONCRETE OR LIQUID GRANITE?

Liquid granite presents a superior alternative to concrete for structural applications, offering higher heat resistance and increased sustainability. Key advantages include reduced structural failure in intense heat and a smaller carbon footprint due to lower cement usage and high-recycled material content.

- **Fire Resistance:** Concrete dehydrates and suffers structural failure under high heat, while liquid granite offers superior endurance.

- **Sustainability:** Liquid granite requires 70% less cement than concrete and incorporates 30–70% recycled materials, reducing CO_2 emissions.

LEARNING OBJECTIVES:

- **1.** Learn to structure two-sample experiments as formal **hypothesis tests**.
- **2.** Compare the **means** of two normal distributions using tests and confidence intervals.
- **3.** Evaluate the **ratio of variances** between two normal distributions.
- **4.** Analyze the **difference in proportions** between two distinct populations.

- **5.** Manage **sample size**, power, and type II error probabilities for statistical accuracy.
- **6.** Use **ANOVA** (Analysis of Variance) to compare means across multiple groups.
- **7.** Utilize **residual plots** to verify if an ANOVA model is appropriate for the data.
- **8.** Apply the **blocking principle** to control for nuisance factors that might skew results.
- **9.** Implement **randomized complete block designs** for effective experimental conduct.

5-1 INTRODUCTION

- **Fig. 5-1** illustrates the extension of statistical inference from a single population to the comparison of two independent populations, focusing on means (μ_1, μ_2) and variances (σ_1^2, σ_2^2). Inferences are based on comparing two separate random samples of sizes n_1 and n_2.